Bangla Regional Dialect Identification: Classical Pipelines versus Transformer Models

CSC 4233 NLP, Section B | Mid-Term Project Proposal (M2) | Group G__ | Members: [Name, ID], [Name, ID], [Name, ID]

1. Problem and use case

Given a short Bangla sentence, predict which regional dialect it is written in (multi-class text classification, 12 classes). Use case: routing and handling regional-language input in Bangla chatbots, call-centre transcripts and speech-to-text pipelines, and measuring how well standard-Bangla tools cope with regional speech. Beyond accuracy, we examine which preprocessing choices help or hurt, because dialect cues often sit in the very words (verb endings, function words) that standard preprocessing removes.

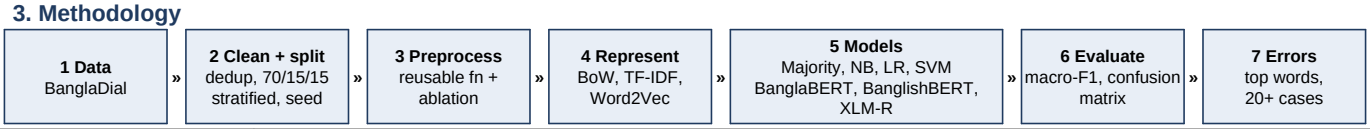
2. Dataset

Main dataset: BanglaDial (Mahi & Khan, Mendeley Data, v2, 2025, DOI 10.17632/sx6ybcps2n.2), **CC BY 4.0** (academic use allowed). 60,729 sentences, 12 classes, mean 6 words per sentence. **Imbalanced**: Kishoreganj 8,691, Chittagong 8,660, Narail 7,743, Rangpur 5,878, Narsingdi 5,735, Tangail 5,410, Standard Bangla 4,403, Barisal 4,046, Sylhet 3,710, Mymensingh 3,096, Noakhali 2,462, Rajshahi 885 (about 10:1), so macro-F1 is the primary metric. The notebook will read the CSV from a public GitHub link (to be added).

Dataset (all Mendeley Data, all CC BY 4.0)	Size / content	Role	Link
BanglaDial	60,729 sent., 12 labels	Main	data.mendeley.com/datasets/sx6ybcps2n
Vashantor	12,500 pairs, 5 dialects, train/val/test	Cross-source check	data.mendeley.com/datasets/bj5jgk878b
ONUBAD	980 sent. x 3 dialects, with standard Bangla	Cross-source check	data.mendeley.com/datasets/6ft99kf89b
Bangla Dialect Dataset	~4,000 pairs, 6 dialects	Cross-source check	data.mendeley.com/datasets/sm63ryv5dt
BD-Dialect	Standard + 5 dialects, aligned	Related / future	data.mendeley.com/datasets/k769s4vk5z
Parallel Corpus (Dialect Bias Eval.)	15,376 pairs, 4 dialects	Related / future	data.mendeley.com/datasets/42gvc83xkg

Our own audit shows BanglaDial already contains about 99% of the sentences of Vashantor, ONUBAD and the Bangla Dialect Dataset, plus 427 repeated rows and 173 sentences carrying more than one label. We therefore deduplicate and drop conflicting labels *before* splitting, and use the other sources only after removing shared sentences. The Bangla Regional Dialects Speech Dataset is not used (restricted access).

3. Methodology



Component	Methods and details
EDA	Class distribution, length statistics, top words per class before/after preprocessing, at least 3 figures.
Preprocessing	One reusable function: regex cleaning (URL, HTML, emoji, digits; we report how rare these are in this corpus), Unicode NFC normalisation, Bangla tokeniser and stop-word list, optional Bangla stemming (limitations discussed).
Ablation	At least 2 variants (stop-words kept vs removed, stemmed vs not, digits kept vs masked); effect on macro-F1.
Representations	Bag of Words; TF-IDF (word unigram vs unigram+bigram, plus character n-grams); averaged Word2Vec (trained on the corpus) or pretrained FastText Bangla vectors.
Classical classifiers	Majority-class baseline; Multinomial Naive Bayes (BoW, TF-IDF); Logistic Regression and Linear SVM on all three representations (Logistic Regression / Gaussian NB on embeddings); one hyper-parameter tuned per model on validation (alpha, C, max_features).
Pre-trained transformers	BanglaBERT, BanglishBERT (Bangla-English bilingual) and XLM-R-base, fine-tuned end-to-end on the same pipeline and splits (Colab GPU, max length 64 since sentences are short; learning rate and epochs tuned on validation). They enter the same comparison table, confusion matrix and error analysis.
Optional	Class-conditional character n-gram language models with smoothing, classified by perplexity (only if the instructor confirms it is in scope).
Protocol	Stratified 70/15/15 split, fixed seed, vectorisers fitted on training data only, test set used once.
Evaluation	Accuracy, macro-precision, macro-recall, macro-F1 in one comparison table; confusion matrix of the best model.
Error analysis	Top-10 informative words per class (Naive Bayes log-probability ratio); at least 20 misclassified examples grouped into error types.
Deliverable	One Colab notebook (Run all) and an IEEE report (max 10 pages).

4. Risks

(i) Very short sentences (about 6 words) limit n-gram and embedding signal. (ii) Label provenance of several BanglaDial district classes is undocumented, so label noise may cap accuracy; we report this honestly. (iii) Strong class imbalance (Rajshahi: 885); handled with stratified splits, macro-F1 and class weights. (iv) Overlap between public dialect datasets can inflate results; handled by deduplication. (v) Limited time before 5 Nov: transformer fine-tuning needs GPU time, so the cross-source check is optional. (vi) The course guideline limits the mid-term to Weeks 1-5 techniques; we request the instructor's approval to include the transformer models.

5. Member roles

#	Member (Name, ID)	Planned role	Notebook / report sections
1	[Name, ID]	Data loading, deduplication, EDA, preprocessing function and ablation	Dataset, EDA (3+ figures), stratified split; regex cleaning, Bangla tokeniser / stop-words / stemming, ablation
2	[Name, ID]	Representations, classifiers, tuning, error analysis	BoW, TF-IDF, Word2Vec; Naive Bayes, LR, SVM; hyper-parameters; informative words, misclassified examples
3	[Name, ID]	Transformer fine-tuning, evaluation, report	BanglaBERT, BanglishBERT, XLM-R; comparison table, confusion matrix; report writing and IEEE formatting

All members write and can explain every notebook section; each member's tasks and percentage contribution are signed off in the report appendix.